HAACKATHON -1 ASSSESSMENT

NAME : POOJA .M

BUSINESS SCENARIO 3
HIGH-VALUE DEPOSIT MONITORING

The bank processes a daily deposit file.

High-value deposits must be flagged for compliance review.

FILE STRUCTURES

1. VSAM ACCOUNT MASTER FILE (KSDS)

Record Layout

Field	Length	Description
Account Number	10	Primary Key
Customer Name	30	Name
Account Type	1	S = Savings, C = Current
Balance	9(7)V99	Numeric
Status	1	A = Active, I = Inactive

Sample VSAM Data

3000000001RAJESH KUMAR S000050000A

3000000002PRIYA SHARMA S000015000A

3000000003ARUN MENON C000200000A

3000000004MEENA IYER S000008000A

3000000005VIKRAM SINGH S000001500I

2. DEPOSIT TRANSACTION FILE (Sequential)

Record Layout

Field	Length
Account Number	10
Deposit Amount	9(7)V99

Sample Transaction Input

3000000001 000020000

3000000002 000150000

3000000005 000010000

3000000011 000005000

BUSINESS RULES

✓ Rule 1 – Account existence

If not found → Failed

✓ Rule 2 – Account status

If inactive → Reject

✓ Rule 3 – Deposit processing

- Add deposit to balance
- If deposit ≥ ₹1,00,000

  → Mark as HIGH VALUE

✓ Rule 4 – Update record

Rewrite VSAM record

SUMMARY REQUIREMENTS

- Total deposits
- Successful deposits
- Failed deposits
- High-value deposits
- Total deposit amount

EXPECTED SUMMARY FORMAT

--------- DEPOSIT BATCH SUMMARY ---------

Total Deposits : XXXX

Successful Deposits : XXXX

Failed Deposits : XXXX

High-Value Deposits : XXXX

Total Deposit Amount : XXXXX

Batch Completed Successfully

JCL REQUIREMENTS

1. IDCAMS KSDS definition
2. Execution JCL to run COBOL program

ERROR HANDLING REQUIREMENTS

File Status	Meaning
00	Success
10	End of file

22	Duplicate key
23	Record not found

Program must not ABEND.

COBOL CODE FILE HANDLING :

********** Top of Data **********

```
IDENTIFICATION DIVISION.
PROGRAM-ID. DEBTAMT.
ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
    SELECT DEP-IN ASSIGN TO DEPIN
      ORGANIZATION IS SEQUENTIAL
      ACCESS MODE IS SEQUENTIAL
      FILE STATUS IS WS-FS-DEP.
    SELECT ACC-MASTER ASSIGN TO ACCVSAM
      ORGANIZATION IS INDEXED
      ACCESS MODE IS DYNAMIC
      RECORD KEY IS ACC-NO
      FILE STATUS IS WS-FS.
DATA DIVISION.
FILE SECTION.
FD DEP-IN.
01 DEPT-REC.
  05 DEPT-ACC-NO    PIC X(10).
  05 DEP-AMT        PIC 9(9).
FD ACC-MASTER.
01 ACC-REC.
  05 ACC-NO        PIC X(10).
  05 ACC-NAME      PIC X(15).
  05 ACC-TYPE      PIC X(1).
```

```
      05 ACC-BAL       PIC 9(9).
      05 ACC-STATUS    PIC X(1).
WORKING-STORAGE SECTION.
01 WS-FS-DEP      PIC X(2).
01 WS-FS          PIC X(2).
01 WS-EOF         PIC X VALUE 'N'.
01 WS-TOTAL       PIC 9(5) VALUE 0.
01 WS-SUCCESS     PIC 9(5) VALUE 0.
01 WS-FAILED      PIC 9(5) VALUE 0.
01 WS-HIGH        PIC 9(5) VALUE 0.
01 WS-TOT-AMT     PIC 9(9)V99 VALUE 0.
01 WS-HIGH-LMT    PIC 9(9)V99 VALUE 100000.
PROCEDURE DIVISION.
000-MAIN-PROCESS.
   PERFORM 100-OPEN-FILES
   IF WS-FS-DEP = '00' AND WS-FS = '00'
      PERFORM 200-PROCESS-DATA UNTIL WS-EOF = 'Y'
   ELSE
      DISPLAY "CRITICAL ERROR:FILE OPEN FAILED"
      DISPLAY "DEPT IN STATUS : " WS-FS-DEP
      DISPLAY "ACC-MASTER STATUS : " WS-FS
      STOP RUN
   END-IF
   PERFORM 300-CLOSE-FILES
   PERFORM 400-DISPLAY-SUMMARY
   STOP RUN.

100-OPEN-FILES.
      OPEN INPUT DEP-IN
      OPEN I-O  ACC-MASTER.
200-PROCESS-DATA.
```

```
        READ DEP-IN
          AT END
             MOVE 'Y' TO WS-EOF
          NOT AT END
             PERFORM 210-PROCESS-DEP
          END-READ.
210-PROCESS-DEP.
         ADD 1 TO WS-TOTAL
         MOVE DEPT-ACC-NO TO ACC-NO
         READ ACC-MASTER
            KEY IS ACC-NO
            INVALID KEY
               DISPLAY "ACCOUNT NOT FOUND : " ACC-NO
               ADD 1 TO WS-FAILED
               EXIT PARAGRAPH
         END-READ
         IF ACC-STATUS = 'I'
             DISPLAY "ACCOUNT INACTIVE : " ACC-NO
             ADD 1 TO WS-FAILED
             EXIT PARAGRAPH
         END-IF
         ADD DEP-AMT TO ACC-BAL
         REWRITE ACC-REC
               INVALID KEY
                 DISPLAY "REWRITE FAILED : " ACC-NO
                 ADD 1 TO WS-FAILED
                 EXIT PARAGRAPH
         END-REWRITE
         IF DEP-AMT >= WS-HIGH-LMT
               ADD 1 TO WS-HIGH
         END-IF
```

```
        ADD DEP-AMT TO WS-TOT-AMT
        ADD 1 TO WS-SUCCESS.
   300-CLOSE-FILES.
        CLOSE DEP-IN ACC-MASTER.
   400-DISPLAY-SUMMARY.
        DISPLAY "------DEPOSIT BATCH SUMMARY----"
        DISPLAY "TOTAL DEPOSITS  : " WS-TOTAL
        DISPLAY "SUCCESSFUL DEPOSITS : " WS-SUCCESS
        DISPLAY "FAILED DEPOSITS : " WS-FAILED
        DISPLAY "HIGH-VALUE DEPOSITS : " WS-HIGH
        DISPLAY "TOTAL DEPOSIT AMOUNT : " WS-TOT-AMT
        DISPLAY "BATCH COMPLETED SUCCESSFULLY".
```

********** Bottom of Data **********

JCL COMPILE CODE

********** Top of Data **********

```
//Z82715E   JOB (123456),'TEST JOB',
//        CLASS=A,MSGCLASS=X,TIME=(0,1)
//*****************************************************
//*             COMPILE THE COBOL PROGRAM
//*****************************************************
//DEBCMP   EXEC PROC=IGYWCL
//SYSLIN   DD DSN=&&LOADSET,UNIT=SYSDA,
//        SPACE=(TRK,(1,1)),DISP=(MOD,PASS)
//SYSIN    DD DSN=Z82715.POOJA.COBOL(DEBTAMT),DISP=SHR
//LKED.SYSLMOD DD DSN=Z82715.POOJA.LOADLIB(DEBTAMT),DISP=SHR
//SYSPRINT DD SYSOUT=*
```

********** Bottom of Data **********

KSDS RUN =DEFINE +LOAD+PRINT

********** Top of Data **********

```
//Z82715R   JOB (1234),'KSDS RUN',NOTIFY=&SYSUID,
//     TIME=(0,1)
//**************KSDS RUN**************
//STEP02    EXEC PGM=IDCAMS
//SYSPRINT  DD SYSOUT=*
//SYSOUT    DD SYSOUT=*
//SYSIN     DD *
    DELETE Z82715.ACCVSAM.KSDS CLUSTER
/*
//************KSDS DEFINE****************
//STEP1     EXEC PGM=IDCAMS
//SYSPRINT  DD SYSOUT=*
//SYSOUT    DD SYSOUT=*
//SYSIN     DD *
    DEFINE CLUSTER (NAME(Z82715.ACCVSAM.KSDS)-
    INDEXED                          -
    RECSZ(80 80)                       -
    KEYS(10 0)                       -
    TRACKS(1 1)                        -
    CISZ(4096)                       -
    VOLUMES(ZXPL01)                      -
    FREESPACE(3 3))                      -
    DATA (NAME(Z82715.ACCVSAM.KSDS.DATA))    -
    INDEX (NAME(Z82715.ACCVSAM.KSDS.INDEX))
//******************KSDS LOAD******************
//STEP3    EXEC PGM=IDCAMS
//SYSPRINT  DD SYSOUT=*
//SYSOUT    DD SYSOUT=*
//IN       DD DSN=Z82715.POOJA.FILE2.DATA,DISP=SHR
//OUT      DD DSN=Z82715.ACCVSAM.KSDS,DISP=SHR
//SYSIN     DD *
```

```
    REPRO INFILE(IN)-
    OUTFILE(OUT)
/*
```

RUN JCL

```
***************************** Top of Data ***************
//Z82715E   JOB (123456),'TEST RUN',
//         CLASS=A,MSGCLASS=X,TIME=(0,1)
//STEP1     EXEC PGM=DEBTAMT
//STEPLIB   DD DSN=Z82715.POOJA.LOADLIB,DISP=SHR
//DEPIN     DD DSN=Z82715.POOJA.FILE1.DATA,DISP=SHR
//ACCVSAM   DD DSN=Z82715.ACCVSAM.KSDS,DISP=SHR
//SYSOUT    DD SYSOUT=*
//SYSPRINT  DD SYSOUT=*
**************************** Bottom of Data *************
```

INPUT SEQUENTIAL FILE

```
* * * * * * * * * * * * * * * * * * * * * * * * * * * *
3 0 0 0 0 0 0 0 0 1 0 0 0 2 0 0 0 0 0 0
3 0 0 0 0 0 0 0 0 2 0 0 0 1 5 0 0 0 0 0
3 0 0 0 0 0 0 0 0 5 0 0 0 1 0 0 0 0 0 0
3 0 0 0 0 0 0 0 1 1 0 1 0 0 0 0 0 0 0 0
* * * * * * * * * * * * * * * * * * * * * * * * * * * *
```

INPUT VSAM FILE

```
*****************************************************  Top of
3000000001RAJESH  KUMAR    S00005000A
3000000002PRIYA  SHARMA    S00001500A
3000000003ARUN  MENON      C00020000A
3000000004MEENA  IYER      S00000800A
3000000005VIKRAMSINGH      S00000150I
*****************************************************  Bottom
```

OUTPUT VSAM -AFTER ADDING DEPOSIT AMOUNT

```
         <====+=====>----+----20---+----3----+-
****** ****   Top of data   ****
000001 3000000001RAJESH  KUMAR    S00007000A
000002 3000000002PRIYA   SHARMA   S00003000A
000003 3000000003ARUN    MENON    C00002000A
000004 3000000004MEENA   IYER     S00000800A
000005 3000000005VIKRAM  SINGH    S00000150I
****** ****   End of data   ****
```

OUTPUT -SDSF SPOOL

```
*****************************************************  TOP
ACCOUNT  INACTIVE  :  3000000005
ACCOUNT  NOT  FOUND  :  3000000011
------DEPOSIT  BATCH  SUMMARY----
TOTAL  DEPOSITS   :  00004
SUCCESSFUL  DEPOSITS  :  00002
FAILED  DEPOSITS  :  00002
HIGH-VALUE  DEPOSITS  :  00000
TOTAL  DEPOSIT  AMOUNT  :  00003500000
BATCH  COMPLETED  SUCCESSFULLY
*****************************************************  BOTT
```